

Mathematics

Section 2

40 Questions

Time Limit: 54 Minutes

Name: _____

Date: _____

MATHEMATICS

This section contains 40 multiple-choice questions. For each question, choose the **best** answer from the four choices provided.

Use the space provided in your test booklet for scratch work. Calculators are not permitted.

Content Areas:

- Numbers and Algebra (approximately 70%)
- Measurement and Data (approximately 30%)

Do not turn to the answer key until you have completed all questions.

1. A nurse is preparing a medication that requires a concentration of 0.5 mg/mL. If the nurse has a vial containing 250 mg of the medication, how many milliliters of solution are needed to provide the exact amount required?
- A) 125 mL
 - B) 250 mL
 - C) 50 mL
 - D) 500 mL
2. A graph plots patient weight (x-axis) vs. medication dose (y-axis). The line has a slope of 0.5 mg/lb. What does this slope represent in context?
- A) The medication dose increases by 0.5 lbs for every 1 mg of patient weight.
 - B) The medication dose increases by 0.5 mg for every 1 lb of patient weight.
 - C) The medication dose increases by 0.5 mg for every 2 lbs of patient weight.
 - D) The medication dose increases by 2 mg for every 1 lb of patient weight.
3. A nurse needs to convert the fraction $\frac{3}{4}$ into a decimal to calculate the correct dosage. Which of the following is the correct decimal equivalent?
- A) 0.34
 - B) 0.075
 - C) 0.43
 - D) 0.75
4. A rectangular basin is 50 cm long, 30 cm wide, and 20 cm deep. How many liters does it hold when full?
- A) 300 L
 - B) 30 L
 - C) 15 L
 - D) 3 L
5. A nurse is preparing a solution by mixing 250 mL of saline with 750 mL of water. What is the ratio of saline to the total volume of the solution?
- A) 1:3
 - B) 3:4
 - C) 1:4
 - D) 1:2

6. Express "2/3 dollars per balloon" with equivalent integer numbers.
- A) \$3 dollars per 9 balloons
 - B) \$2 dollars per 3 balloons
 - C) \$3 dollars per 6 balloons
 - D) \$1 dollar per 9 balloons
7. An L-shaped counter is made from a 6 ft by 4 ft rectangle with a 2 ft by 2 ft corner removed. What is the area?
- A) 22 ft²
 - B) 28 ft²
 - C) 24 ft²
 - D) 20 ft²
8. A patient is prescribed 0.75 mg of a medication. If the available tablets are scored and contain $\frac{1}{4}$ mg each, how many tablets should the nurse administer?
- A) $\frac{3}{4}$ tablet
 - B) 6 tablets
 - C) 2.25 tablets
 - D) 3 tablets
9. A patient is 5 feet 4 inches tall. Convert to centimeters using 1 inch equals 2.54 cm.
- A) 162.6 cm
 - B) 167.6 cm
 - C) 165.1 cm
 - D) 160.0 cm
10. Which of the following expressions is equivalent to $3(x + 4)$?
- A) $3x + 7$
 - B) $3x + 12$
 - C) $3x + 4$
 - D) $3x + 16$
11. A circle graph shows budget allocation: Salaries 35%, Supplies 20%, Marketing 15%, Maintenance 10%, Training 20%. If the total budget is \$120,000, how much more is spent on salaries than on training?
- A) \$12,000
 - B) \$30,000
 - C) \$18,000
 - D) \$24,000

12. Emergency department wait times in minutes are: 15, 22, 18, 25, 20, 30, 18. What is the median wait time?
- A) 18
 - B) 25
 - C) 20
 - D) 22
13. Convert 2.4 kilometers to centimeters.
- A) 24,000 cm
 - B) 2,400,000 cm
 - C) 24,000,000 cm
 - D) 240,000 cm
14. A graph shows sales increasing from 200 to 260. What is the increase?
- A) 60
 - B) 50
 - C) 70
 - D) 40
15. A nurse is calculating the dosage for a patient. If 'x' represents the patient's weight in kilograms, which algebraic expression represents 'three times the weight plus five milligrams of medication'?
- A) $3x - 5$
 - B) $x + 3 + 5$
 - C) $3(x + 5)$
 - D) $3x + 5$
16. Medication errors per 1000 doses by quarter: Q1 4.5, Q2 3.6, Q3 2.7, Q4 1.8. What is the average decrease per quarter?
- A) 0.7
 - B) 0.9
 - C) 1.0
 - D) 0.8
17. A doctor prescribes 5 grams of a medication. How many milligrams is this?
- A) 5000
 - B) 50000
 - C) 500
 - D) 50

18. A nurse is preparing a medication dose using the formula: $\text{Dose} = (\text{Weight} \times 2.5) - 10$. If a patient weighs 6 kg, how much medication (in grams) should be administered? Round to the nearest tenth if necessary.

- A) 2 grams
- B) 10 grams
- C) 5 grams
- D) 8 grams

19. A nurse is calculating the total dosage of a medication for a patient who receives 250 mg of drug A, 150 mg of drug B, and 50 mg of drug C in a single administration. What is the total amount of medication in milligrams?

- A) 300 mg
- B) 400 mg
- C) 350 mg
- D) 450 mg

20. A scatter plot displays study hours versus test scores for a group of students. The plotted points generally slope upwards from left to right. What does this trend indicate?

- A) Test scores are constant regardless of the number of study hours.
- B) There is no relationship between study hours and test scores.
- C) As study hours increase, test scores tend to increase.
- D) As study hours increase, test scores tend to decrease.

21. What is the perimeter of an equilateral triangle with each side measuring 4 cm?

- A) 16 cm
- B) 48 cm
- C) 4 cm
- D) 12 cm

22. A nurse is preparing a medication solution by mixing $\frac{3}{4}$ teaspoon of liquid medication into a container that already holds $\frac{1}{2}$ teaspoon of water. What is the total volume of the mixture in teaspoons?

- A) $1\frac{1}{4}$
- B) $\frac{5}{4}$
- C) $\frac{7}{4}$
- D) $1\frac{1}{2}$

23. A hospital staffing table shows: Nurse A works three 8-hour shifts, and Nurse B works four 6-hour shifts. What is the total number of hours worked by both nurses?
- A) 64 hours
 - B) 48 hours
 - C) 56 hours
 - D) 32 hours
24. Which of the following is the most appropriate unit for measuring the weight of a vitamin tablet?
- A) Liters
 - B) Milligrams
 - C) Kilograms
 - D) Grams
25. A patient is prescribed a medication dose of 250 mg. The available supply of the medication is in 5 mg tablets. How many tablets should the nurse administer?
- A) 5 tablets
 - B) 125 tablets
 - C) 25 tablets
 - D) 50 tablets
26. A histogram shows the distribution of test scores. The interval 70-79 has a frequency of 15, and 80-89 has a frequency of 25. If a student is chosen at random from the combined group of these two intervals, what is the probability they scored 80 or above?
- A) $\frac{25}{40}$
 - B) $\frac{15}{40}$
 - C) $\frac{25}{15}$
 - D) $\frac{5}{8}$
27. A table shows the number of apples sold per day: Mon=15, Tue=20, Wed=18, Thu=22, Fri=25. What is the average number of apples sold per day?
- A) 21
 - B) 19
 - C) 18
 - D) 20

28. Add $\frac{3}{5} + \frac{1}{3} = ?$

A) $\frac{17}{15}$

B) $\frac{16}{15}$

C) $\frac{8}{15}$

D) $\frac{14}{15}$

29. A patient weighs 70 kg. Convert to pounds using 1 kg equals 2.2 lb.

A) 144 lb

B) 150 lb

C) 160 lb

D) 154 lb

30. A client has a temperature of 39.5°C. The standard conversion formula is $F = (C \times \frac{9}{5}) + 32$. What is the temperature in Fahrenheit?

A) 104.1°F

B) 102.1°F

C) 101.1°F

D) 103.1°F

31. A scatter plot shows data points that cluster tightly around a line sloping downward. The correlation coefficient (r) is calculated. Which value best represents this relationship?

A) -0.95

B) 0.95

C) 0.20

D) -0.20

32. A nurse is calculating the cost of medication for a patient. The prescription requires 3 bottles of a drug, and each bottle costs \$15.50. Which algebraic expression represents the total cost, where x represents the number of bottles?

A) $x / 3$

B) $15.50x$

C) $3 + 15.50$

D) $15.50 * 3$

33. A runner runs 5 kilometers in 25 minutes. What is their average speed in meters per minute?
- A) 5,000 m/min
 - B) 200 m/min
 - C) 100 m/min
 - D) 125 m/min
34. A nurse is preparing a medication where the dosage is calculated using the inequality $3x + 5 > 20$. What is the minimum whole number of units (x) the nurse must use to ensure the dosage meets the requirement?
- A) 6
 - B) 4
 - C) 5
 - D) 7
35. What is the area of a circular IV site with a radius of 5 cm? Use $\pi \approx 3.14$.
- A) 78.5 cm²
 - B) 157 cm²
 - C) 31.4 cm²
 - D) 25 cm²
36. A cylindrical tank has a radius of 3 feet and a height of 10 feet. What is the approximate volume? (Use $\pi \approx 3.14$).
- A) 30 cubic ft
 - B) 94.2 cubic ft
 - C) 282.6 cubic ft
 - D) 90 cubic ft
37. A table shows daily patient admissions: Monday=12, Tuesday=15, Wednesday=14, Thursday=20, Friday=18. Which day had the greatest number of admissions?
- A) Tuesday
 - B) Thursday
 - C) Monday
 - D) Friday
38. A bar graph displays a patient's temperature readings over four days: Monday: 98.6°F, Tuesday: 99.2°F, Wednesday: 100.1°F, Thursday: 99.8°F. On which day was the patient's temperature the highest?
- A) Monday
 - B) Thursday
 - C) Tuesday
 - D) Wednesday

39. A nurse is calculating the cost of a medication that costs \$15 per unit. If the nurse needs to purchase x units, which algebraic expression represents the total cost?

A) $15 + x$

B) $15 - x$

C) $x / 15$

D) $15x$

40. A box has volume 100 and base area 20. What is its height?

A) 5

B) 6

C) 4

D) 7

ANSWER KEY

Mathematics — 40 Questions

1. D	9. A	17. B	25. D	33. B
2. C	10. B	18. C	26. D	34. C
3. D	11. C	19. D	27. D	35. A
4. B	12. C	20. C	28. D	36. C
5. C	13. C	21. D	29. D	37. B
6. A	14. A	22. A	30. D	38. D
7. D	15. D	23. A	31. A	39. D
8. D	16. B	24. B	32. B	40. A